

Register Lens

Your register already knows.
Now DVDMS will too.

Smart Health & Supply Chain Resilience



The notebook is right. The screen is out of date.

Busy OPD, no time to type twice.

Auditors record that DVDMS is not updated regularly because of the rush of work and shortage of staff.

So DVDMS drifts from reality.

Stock differs from the registers, and dispensed medicines can even show as expired.

Everything built on DVDMS inherits it.

Dashboards and forecasts are only as good as the data underneath.

₹14.52 crore

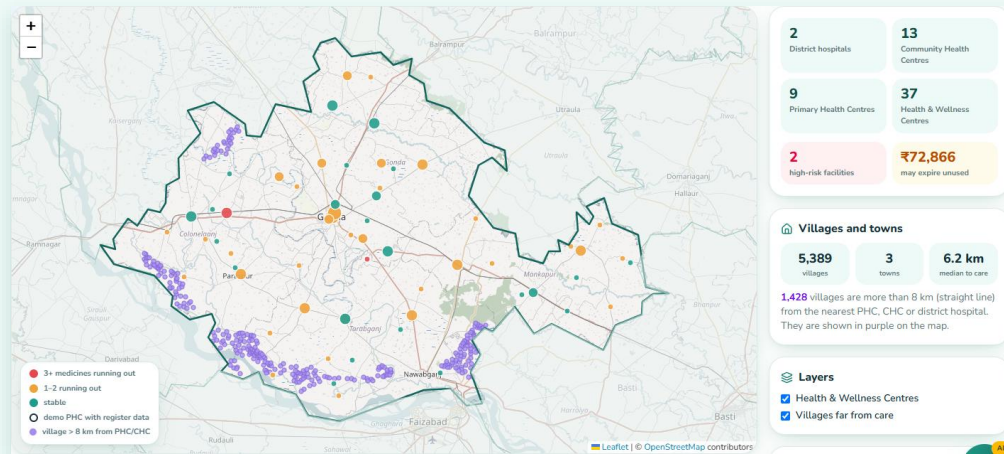
expired drugs reported lying in Haryana

₹23.03 crore

expired drugs reported in Karnataka, 2019–22



Beyond one PHC: all of Uttar Pradesh



Gonda: every facility at its real location, coloured by stock risk; purple = villages more than 8 km from a PHC/CHC (1,428 in Gonda)

3,948 public health facilities mapped
(OpenStreetMap)

75 districts, with villages far from care

84 expiry transfers suggested in Barabanki alone

Phase 3, live: batches that would expire unused are matched to the nearest facility running short; the district officer approves each move and downloads transfer orders.

One photo. Four steps. Zero extra typing.



Snap

Photograph the day's register page.



Read

Gemini reads Hindi + English handwriting and scores each cell.



Confirm

The pharmacist checks only flagged cells.



Sync

DVDMS bulk entry + Hindi/English district alert.

No new hardware · no new forms · the pharmacist always approves

A real handwritten page, end to end

What Gemini saw

Original photo

gemini unsure Numbers don't add up Batch check vs DVDMS

Tap any cell to fix it, or confirm the line as it is.

MEDICINE	BATCH	EXPIRY	OPEN	RECV	ISSUED	CLOSE
Amoxicillin 500 mg	AM3038	2027-06	600	0	72	528
amox 500mg → Issued and Filled, Amox 500 mg = 100 mg						
Metformin 500 mg	MF2219	2027-07	900	0	140	760
→ Metformin 500 mg						
Metformin 500 mg	AL5530	2027-12	700	0	95	605
→ Metformin 500 mg						
Cefixime 10 mg	CT1187	2026-11	350	0	40	310
→ Cefixime 10 mg						
Zinc 20	ZN6802	2028-04	300	0	25	275
→ Zinc 20mg 20 mg # Review this cell and see 280 - 10 = 270, but change to 285						
Aggravate	MT4471	2028-02	400	0	80	340
→ Metformin 400 mg						
Bu 400	BB2296	2027-10	500	0	45	455
→ Metformin 400 mg						

Start over Confirm all (1) Compare with DVDMS

Review: the wrong Zinc total is caught in red

DVDMS vs the register - PHC Banki

Running out (≤ 7 days) Getting low (≤ 15 days) DVDMS overstated

Medicine	DVDMS status	DVDMS stock	Reality
Antidopine 5 mg	Running out	3,050 tabs	112.1 d
Metformin 500 mg	Running out	6,158 tabs	58.3 d
Metronidazole 400 mg	Running out	3,158 tabs	52 d
Oral Rehydration Salts WHO formula	Running out	1,032 sachets	202 sachets
Paracetamol 500 mg	Getting low	18,525 tabs	1,515 tabs
Amoxicillin 500 mg	Getting low	6,450 caps	528 caps

Compare: register vs DVDMS days of stock

District alert

English

DVDMS bulk-entry file

All 10 confirmed lines in a ready-to-upload sheet: facility, drug code, batch, expiry, opening, received, issued and closing stock.

Download Excel file

1 batch expire within 60 days Phase 3

Send this to a nearby facility that urgently needs them, with the district officer's approval. Cefixime.

Plan transfers

Sync: DVDMS entry + alert in Urdu, Hindi, English

Live end to end: our own handwritten page in Hindi + English, with Hindi numerals and a crossed-out value. Gemini read all 10 lines correctly, the planted Zinc error was caught, and 54 → 45 was understood as a correction.

React + FastAPI + Gemini API on Cloud Run · works on PC and as an installable phone app

Where AI works, and where plain code does

Photo phone camera

OpenCV clean-up flatten page, remove shadows, boost ink

Gemini reads handwriting to JSON + confidence

Code checks match drugs, balances, DVDMS drift, days of stock

Gemini writes district alert, Hindi + English

The pharmacist confirms between “reads” and “checks”. Nothing is uploaded without a human yes.

1

Structured output

Gemini returns schema-checked JSON with a confidence score per cell.

2

Numbers are never generated

Drift and days of stock are computed by checkable code.

3

Resilient to limits

Fails over across Gemini models, skips exhausted ones, lighter models for alerts, stored readings for demo pages.

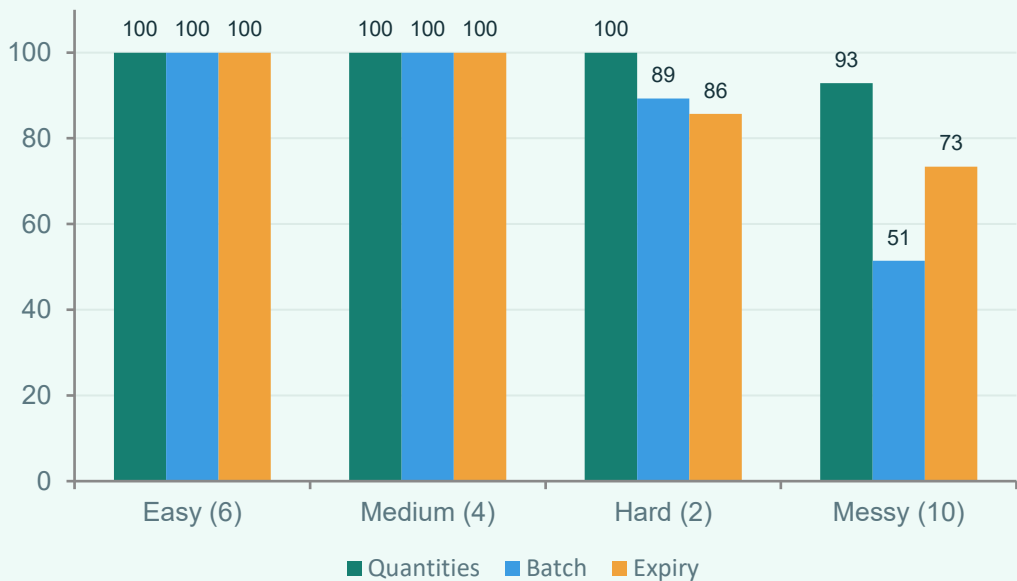
4

Names in two languages

PCM, Paracetamol and पैरासिटामोल resolve to one medicine.

Does it work? Measured, with the misses.

Cell accuracy by page type (%), 22 synthetic pages



Drug names: 100% everywhere

Quantities, which drive stock-out numbers: 100% on clean pages, 93% on messy ones.

The weak spot

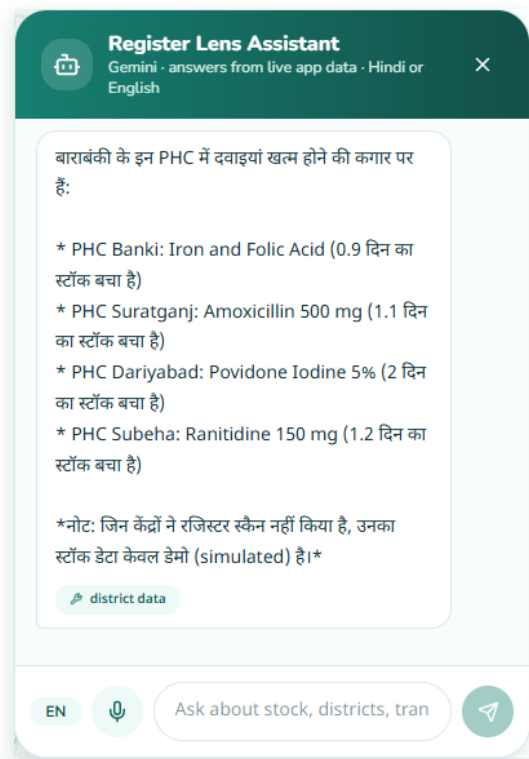
Small cursive batch codes and expiry dates on messy pages (G/U, E/C, 3/8). Gemini's own confidence caught only 4 of 104 wrong values.

Our fix

DVDMS batch cross-check + balance check flagged 63 of 74 lines with errors for the pharmacist, 0 false alarms.

Messy pages: ruled notebooks, no column lines, cursive, corrections, Hindi numerals, stains, folds, blur. All pages synthetic; the demo DVDMS is a mock seeded with correct batches, so the cross-check shows the mechanism, not a field catch rate.

Ask it anything. Use it on any phone.

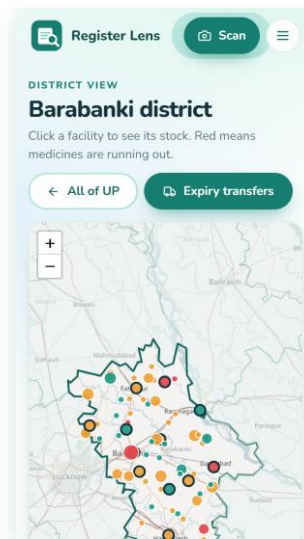
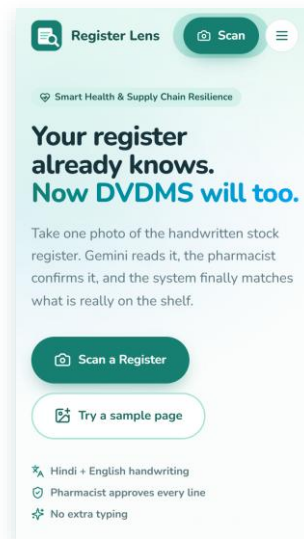


Gemini assistant with function calling

Looks up districts, facilities, nearby stock and transfer plans before answering, so it never invents a number. Replies in Hindi, Hinglish or English. Voice input.

Installable phone app (PWA)

Open the link in Chrome, tap Install. Own icon, full screen, Scan opens the camera. No app store needed.



Phone views: home and district map

Safeguards: unsure cells yellow • balance errors red • batch cross-check • nothing uploaded automatically • officer approves every transfer.

Pilot-ready in four weeks

Week 1

Collect real register photos from 3–5 PHCs in one district. Measure accuracy.

Week 2

Map each register template. Match DVDMS bulk-entry fields with the state IT team.

Week 3

Shadow mode: run beside the current process and compare stock numbers.

Week 4

Go live with pharmacists, district officer sign-off on alerts.

What it needs: a phone camera, a browser, a Gemini API key. It reads the register that already exists and outputs a bulk-entry file for the existing system, so there is no new hardware and no new workflow to learn.

Built to travel across states and BRICS nations



Language

Phase 2, live: district alerts in 16 Indian languages; UP defaults to Hindi, Urdu, English.

हिन्दी اردو भोजपुरी தமிழ் తెలుగు
বাংলা मराठी ଓଡ଼ିଆ ಕನ್ನಡ



Register templates

Layout differs by state; one template file per register lets Gemini read a new format without retraining.

column names · units · date style



Medicine catalogue

Each country brings its own essential-medicines list and local aliases.

NLEM (India) → other national lists

Already statewide: 3,948 facilities in all 75 UP districts are on the map today. A new state needs its facility list, register template and languages, not new code.

How this maps to the judging criteria

Criterion	Weight	Evidence you can open
AI / Technical execution	25%	Gemini reads handwriting to JSON; function-calling assistant; multilingual alerts; OpenCV clean-up; 22-page evaluation
Problem–solution fit	20%	Targets the audit-documented gap: DVDMS vs registers, at the pharmacist's end-of-day moment
Depth & reach across India	20%	All 75 UP districts, 3,948 facilities, villages far from care; 16 languages; config-driven for other states
Deployability & scalability	20%	Runs on Cloud Run; installable phone app; DVDMS bulk-entry output; four-week pilot plan
Impact potential	15%	Stock-outs and expiry losses (reported in crores) at the source; expiry transfers with officer approval

Deliverables: source code repository · demo video (3–5 min) · this deck · live prototype link.

Your register already knows. Now DVDMS will too.

One photo a day keeps the district's numbers true.

